

EDUCATION

Masters of Computer Science, Dalhousie University
Bachelor of Technology, Electronics and Communication, GGSIPU University

Jan 2021 — Dec 2022
Aug 2012 — May 2016

SKILLS

Programming	Python, JavaScript
Web Technologies	JavaScript, HTML, CSS, jQuery
Cloud Technologies	AWS Lambdas, S3, SageMaker, Bedrock, Kubernetes, Docker
BI tools	Alteryx, DOMO, QlikSense, Tableau, PowerBI, Excel
Frameworks	Django, Flask, Kafka, Rabbit MQ
Project Management	Git, CI/CD, Agile, Scrum, GitHub, GitLab
Machine Learning	PyTorch, Sci-kit learn, Pandas, NumPy, Transformers, Spacy, Descriptive Statistics, Hypothesis Testing
Generative AI	Openai API, LangChain, HuggingFace Trainer API, Instruct Finetuning, Langflow, LangSmith, Gradio Application, GPT 4 Evaluation, Lora/QLora Training, Prompt Engineering, Mistral AI, Chain-of-Thought Framework, RAG Framework, Vector Database (Chroma DB and Pinecone)

EXPERIENCE

Senior ML/AI Engineer
Behavox

Jan 2023 — Current
Montreal, Canada

- Collaborated with the R&D team to develop an AI platform powered by large language models (LLMs) tailored for the finance compliance domain.
- Pretrained LLMs on proprietary financial chat datasets using Masked Language Modeling on 2B tokens, optimized for multi-GPU (H100) environments with Python's Pytorch, transformers and Accelerate module.
- Built a Chat summarization tool within the AI platform by implementing prompt-based instruction fine-tuning on SME-curated datasets using ALPACA format
- Developed a multilingual keyword extraction tool using the KeyBERT framework and a custom SBERT model to automate keyword extraction for risk policies like collusion, Anti-Money laundering.
- Curated and implemented fraud detection solution, combining keyword-based and machine-learning models for 40+ compliance policies, to top banks in Canada like Scotia and BMO, resulting in a \$500K revenue boost within a quarter and serving as a gateway to the Behavox compliance solution suite.
- Developed and automated a training pipeline for a communication surveillance product, incorporating logistic regression, BERT-family models, ONNX-format models, and LLM-based text classification. Achieved a 90% reduction in manual release efforts by streamlining the processes for training, validation, and deployment.
- Leveraged cloud technologies such as AWS Lambda, S3, Docker, and GitLab CI/CD, along with tools like Poetry, SlackBot, and the Hugging Face library, to design a robust system. The pipeline supports automated updates triggered by data ingestion and provides researchers with the flexibility to adjust parameters, enabling performance optimization.
- Improved training outcomes by implementing checkpoint selection and FPR-based threshold optimization, quantization and converting into CPU-compatible format for multilingual DeBERTa and SBERT model designed to detect bad actors across 40+ compliance policies.

ML research Intern
Mitacs

Aug 2022 — Apr 2023
Halifax, Canada

- Developed and deployed a real-time human-in-the-loop machine learning framework for classifying multi-page scanned documents in the Real-estate domain. Integrated OCR, NLP, and machine learning techniques for preprocessing, vectorization, and modelling to ensure accurate and efficient document classification.
- Fine-tuned and optimized diverse Transformer architectures, such as BERT, TransformerXL, Longformer, and RoBERTa XLM. Additionally, developed a lightweight Bag-of-Words (BoW) with the Naive Bayes model, achieving a 98% F1 score and ensuring compatibility with low-hardware systems on S3. Successfully deployed the solution on AWS SageMaker for scalable and efficient performance.
- Conducted extensive experiments and in-depth analysis, culminating in an original thesis focused on document classification and information extraction. Delivered novel insights and methodologies, advancing the field of automated document processing.

Teacher Assistant
Dalhousie University

Jan 2021 — Dec 2022
Halifax, Canada

- Assisted in delivering lab-based courses for machine learning, data science, and statistical methods across three academic terms.

- Taught machine learning and deep learning concepts with Python, covering topics such as classification algorithms (Decision Tree, SVM, KNN), neural networks with Keras, generative models, clustering, and data simulations.
- Designed and conducted lab tutorials and quizzes on Python programming, data preprocessing, and applied statistical tests, including t-Test, Chi-Square Test, ANOVA, paired sample t-Test, and z-Test.
- Developed assignments for data science processes, focusing on classification and clustering models for diverse data types (numerical, geospatial, time series, and text).
- Curated and prepared materials to ensure clarity and engagement for students in both undergraduate and graduate-level courses.

Senior Business Analyst

Feb 2019 — Jan 2021

Novartis Pharmaceutical

Hyderabad, India

- Designed and deployed a web-based application using the Flask framework for the marketing team to simulate and optimize promotional budgets. Utilized sigmoidal response curves and market mix model results, resulting in data-driven budget allocation and improved campaign effectiveness.
- Used NLP techniques such as keyword extraction, sentiment analysis, clustering, and topic modelling to analyze social media data from the brand's Facebook page, Twitter posts, and websites. The insights helped address issues in the company's patient programs.
- Collaborated with the ZS team to create a Python-based parser to extract text and metadata from XML files generated by PPT and PDF documents. Streamlined data cleaning using Pandas string functions and NLTK, enabling efficient text clustering for insights.
- Conducted A/B testing to assess the new strategy employed by the China marketing team for Cosyntex drug sales. Utilized T-tests to reveal significant improvements in rep sales between two quarters, validating the approach's effectiveness.
- Led end-to-end client engagements, contributing to wireframe designs and data modelling using Alteryx ETL tools. Built scalable dashboards in QlikSense for multiple drugs and therapy areas, improving stakeholder decision-making through actionable insights.

Data Analyst

Mar 2018 — Jan 2019

Decision Tree Analytics and Services

Gurugram, India

- Developed interactive digital marketing dashboards in DOMO and Tableau, improving UI/UX through storyboarding techniques and showcasing performance KPIs for clients
- Worked on QA automation using Selenium, saving 3 hours of daily manual tasks and enhancing reliability
- Extracted and processed large datasets from social media platforms (Google Ads, Facebook Insights, Yahoo Gemini, App Leads, Taboola Ads) using Python and RESTFUL APIs

Associate Software Developer

Jan 2017 — Mar 2018

Accenture

Gurugram, India

- Implemented and maintained Informatica PowerCenter ETL workflows, optimizing storage and processing of historical data in a data warehouse with a snowflake schema
- Enhanced ETL flow efficiency by replacing complex data processing with SQL queries at the source, resulting in a significant 90% reduction in runtime
- Developed a chatbot prototype using the naïve Bayes classification algorithm, winning a cash prize during the innovation month and addressing team project queries

PROJECTS

RAG implementation with Phi-3.5B RAG, Python, llama Index, ChromaDB

- Designed and developed a pipeline for context retrieval and interactive querying using a custom LLM (Phi-ONNX) and HuggingFace embeddings. Built and stored a vector-based index from PDF documents for efficient retrieval.
- Utilized HuggingFace's embedding models and Microsoft Phi-ONNX to generate contextual responses. Designed an interactive querying interface with similarity postprocessing and dynamic prompt generation for accurate response synthesis

LLM Model Interfaces, Python, HuggingFace, ONNX Models, ML pipeline design

- This project is a Python application designed for machine learning model training and inference. It leverages models in both Hugging Face and ONNX formats, enabling efficient training and inference processes. The application is modular, with each Python file serving a specific purpose.